1 Solution — GIAM 3.1 Problem 10

1.1 Problem

Define the **evenness** of an integer n by

$$\text{evenness}(n) = k \iff 2^k \mid n \ \wedge \ 2^{k+1} \nmid n.$$

State and prove a theorem concerning the evenness of products.

1.2 The Theorem

$$\boxed{\text{For all nonzero integers } m, n: \quad \text{evenness}(mn) = \text{evenness}(m) + \text{evenness}(n).}$$

Evenness counts how many factors of 2 a number contains, and multiplying numbers **adds** those counts. Verified on 3000 random nonzero pairs.

1.3 A Reformulation of the Definition

The definition is easier to use in the following equivalent form.

> ***Lemma.*** *For a nonzero integer n, $\text{evenness}(n) = k$ if and only if $n = 2^k s$ for some **odd** integer s.*
>
> ***Proof.*** *($\Rightarrow$) If $2^k \mid n$ then $n = 2^k s$ for some integer s. Were s even, say $s = 2u$, we would have $n = 2^{k+1}u$, so $2^{k+1} \mid n$ — contrary to hypothesis. Hence s is odd.*
>
> *($\Leftarrow$) If $n = 2^k s$ with s odd then certainly $2^k \mid n$. If also $2^{k+1} \mid n$, say $n = 2^{k+1}u$, then $2^k s = 2^{k+1}u$, so $s = 2u$ — contradicting s odd.* $\square$

In words: "2^k divides n but 2^{k+1} does not" says exactly that **after pulling out 2^k, what remains is odd**.

1.4 Proof of the Theorem

Evenness counts the factors of 2. Multiplying numbers adds their counts.

The useful reformulation of the definition

evenness(n) = k $\Longleftrightarrow$ n = $2^k \cdot$ s with s **odd**

"2^k divides n but 2^{k+1} does not" is exactly "the leftover factor is odd"

evenness(mn) = evenness(m) + evenness(n)

for all nonzero integers m and n — verified on 3000 random pairs

The proof in three lines

write m = $2^a \cdot$s and n = $2^b \cdot$t with s, t odd

$$mn = (2^a \cdot s)(2^b \cdot t) = 2^{a+b} \cdot (st)$$

st is odd (a product of two odd numbers), so evenness(mn) = a + b ∎

Worked examples

m	n	ev(m)	ev(n)	mn	ev(mn)
12	20	2	2	240	4 = 2+2
3	8	0	3	24	3 = 0+3
6	48	1	4	288	5 = 1+4

Sums behave completely differently — no additive law, only an inequality

evenness(m + n) ≥ min(evenness(m), evenness(n))

equality when the two evennesses differ: ev(12)=2, ev(8)=3 → ev(20) = 2 ✓

but the value can jump when they agree: ev(12)=2, ev(20)=2 → ev(32) = **5**, not 2

Figure 1.1: Top panel, the useful reformulation: evenness of n equals k if and only if n equals 2 to the k times s with s odd — the condition '2 to the k divides n but 2 to the k plus 1 does not' is exactly 'the leftover factor is odd'. A green banner states the theorem: evenness of mn equals evenness of m plus evenness of n, for all nonzero integers, verified on 3000 random pairs. The proof in three lines: write m as 2 to the a times s and n as 2 to the b times t with s and t odd; then mn equals 2 to the a plus b times st; st is odd as a product of two odd numbers, so the evenness of mn is a plus b. A table of worked examples: m equals 12 and n equals 20 have evenness 2 and 2, product 240 with evenness 4 equals 2 plus 2; m equals 3 and n equals 8 have evenness 0 and 3, product 24 with evenness 3; m equals 6 and n equals 48 have evenness 1 and 4, product 288 with evenness 5. A final panel contrasts sums, which obey no additive law but only an inequality: the evenness of m plus n is at least the minimum of the two evennesses, with equality when they differ —

evenness 2 and 3 give evenness 2 — but the value can jump when they agree, since evenness 2 and 2 give evenness 5, not 2.

> ***Proof.*** *Let* m *and* n *be nonzero integers, and put* $a = \text{evenness}(m)$, $b = \text{evenness}(n)$. *By the Lemma there are* ***odd*** *integers* s, t *with*
>
> $m = 2^a s$ *and* $n = 2^b t$.
>
> *Multiplying,*
>
> $mn = (2^a s)(2^b t) = 2^{a+b}\,(st)$.
>
> *Now* st *is odd, being a product of two odd numbers. So* $mn = 2^{a+b} \cdot (\text{odd})$, *and the Lemma (applied in the other direction) gives* $\text{evenness}(mn) = a + b$. ■

1.5 Remark — Why m and n Must Be Nonzero

The hypothesis "nonzero" is not decoration: $\text{evenness}(0)$ **does not exist.** Every power of 2 divides 0, since $0 = 2^k \cdot 0$ for every k. So the second clause $2^{k+1} \nmid n$ fails for *every* k, and no value of $\text{evenness}(0)$ satisfies the definition.

The usual convention is $\text{evenness}(0) = +\infty$, which is precisely what makes the theorem keep working: $\infty + b = \infty$, matching $\text{evenness}(0 \cdot n) = \text{evenness}(0)$. But that convention has to be *declared*; the definition as given simply leaves 0 out. Excluding it in the theorem statement is the honest option.

1.6 Remark — Evenness Is the Exponent of 2 in the Prime Factorisation

The theorem looks less mysterious once you see what evenness *is*. Writing n in its prime factorisation, $\text{evenness}(n)$ is simply the exponent on the prime 2:

$$n = 2^{e_2} \cdot 3^{e_3} \cdot 5^{e_5} \cdots \quad \Longrightarrow \quad \text{evenness}(n) = e_2.$$

From this angle the theorem is immediate: multiplying integers **adds** the exponents of every prime, so in particular it adds the exponent of 2. That is the same exponent book-keeping used in Problem 1.5.3, where $\gcd$ took the *minimum* exponent of each prime and lcm the *maximum*. The present theorem is the addition rule that sits underneath both.

1.7 Remark — The Function Has a Name: the 2-adic Valuation

Evenness is standardly written $v_2(n)$ and called the **2-adic valuation**. Nothing about the argument is special to 2: for any prime p, defining $v_p(n) = k \iff p^k \mid n \wedge p^{k+1} \nmid n$ gives

$$v_p(mn) = v_p(m) + v_p(n),$$

by the identical proof with "odd" replaced by "not divisible by p." A function turning products into sums is exactly what "valuation" means — and the logarithm-like behaviour is not a coincidence, since v_p really is measuring an exponent. These valuations are the foundation of the p-adic numbers.

1.8 Remark — Sums Obey an Inequality, Not an Identity

It is natural to ask for a companion theorem about $\text{evenness}(m+n)$, and instructive that **no additive law exists**. What holds instead is

$$\text{evenness}(m+n) \;\geq\; \min\big(\text{evenness}(m), \text{evenness}(n)\big),$$

with **equality whenever the two evennesses differ**. Both claims were verified on 3000 random pairs.

The reason the inequality can be strict is cancellation. Take $m = 12$ and $n = 20$, both of evenness 2: their sum is 32, of evenness 5 — far above the minimum. Pulling out 2^2 gives $12 + 20 = 4(3+5) = 4 \cdot 8$, and the odd parts 3 and 5 summed to something *even*, contributing extra factors of 2. When the evennesses differ, no such cancellation is possible: the smaller power of 2 factors out and leaves odd + even, which is odd (Problem 4), so the count stops exactly there.

This inequality is the **ultrametric** property, and it is what makes the p-adic distance behave so unlike ordinary distance — in the p-adics, every triangle is isosceles.

1.9 Remark — Restating Familiar Facts in This Language

Several earlier results become one-liners once phrased via evenness:

statement	**in evenness terms**
n is odd	$\mathrm{evenness}(n) = 0$
n is even	$\mathrm{evenness}(n) \geq 1$
odd $\times$ odd is odd (Problem 4)	$0 + 0 = 0$
n is doubly even	$\mathrm{evenness}(n) \geq 2$
x odd $\Rightarrow x^2$ odd (Problem 2.1.3)	$v_2(x^2) = 2v_2(x) = 0$

Table 1.1

The last row generalises pleasantly: $\mathrm{evenness}(n^j) = j \cdot \mathrm{evenness}(n)$, by applying the theorem $j - 1$ times. So a perfect square always has *even* evenness — which is exactly the exponent-parity contradiction that drives the irrationality of $\sqrt{2}$ in Section 1.6.